

CISCO SECURE WORKLOAD

CISA Zero Trust Maturity Model

Zero Trust Maturity Model v2.0 — CSW Pillar Mapping

Prepared for: [Customer Name]

Prepared by: Cisco Systems, Inc.

Date: [Month Year] | Version 1.0 | Cisco Confidential

**Regulatory Standard: CISA Zero Trust Maturity
Model v2.0 (2023)**

Environment: Hybrid (On-Premises + Cloud)

Deployment Stage: Fresh Installation

How to read the Coverage column

Direct means Secure Workload produces the primary evidence artifact for that row — segmentation, policy, flow and process telemetry, inventory, vulnerability and reachability, or forensic reconstruction. Direct is still evidence, not an attestation.

Supporting means Secure Workload contributes an input that feeds or complements another control, tool, or process — detection input only, reconciled against an external register, or one log source among several.

Evidence Required means Secure Workload has no record for that row. The customer supplies the contract, the analysis, or the HR / physical-security evidence from another control.

Every report in this set now uses the same three coverage words — Direct, Supporting, and Evidence Required. Coverage describes what evidence Secure Workload produces; it never means the control is passed.

Executive Summary

The CISA Zero Trust Maturity Model (ZTMM) v2.0 defines maturity across five pillars: Identity, Devices, Networks, Applications & Workloads, and Data. Each pillar progresses through four stages: Traditional → Initial → Advanced → Optimal. Cisco Secure Workload (CSW) is the primary platform for achieving Advanced-to-Optimal maturity in the Networks pillar and significant progress in the Applications & Workloads and Data pillars.

CSW's workload-level micro-segmentation, continuous ADM, and forensic telemetry align directly with CISA's emphasis on continuous validation, least-privilege access, and dynamic policy enforcement — the hallmarks of Optimal maturity.

Pillar Coverage & Maturity Achievable

Pillar	CSW Role	Maturity Achievable	Key CSW Capability	Coverage
Networks	Primary	Advanced → Optimal	Micro-segmentation, encryption enforcement, continuous ADM	Direct
Applications & Workloads	Primary	Advanced → Optimal	Workload identity, least-privilege policy, vulnerability management	Direct
Data	Supporting	Initial → Advanced	Data flow visibility, encryption enforcement, access logging	Direct
Identity	Supporting	Initial → Advanced	Identity-verified access path enforcement via policy	Supporting
Devices	Supporting	Initial → Advanced	OS fingerprint + process hash workload identity	Supporting

Maturity Stage Mapping by Pillar

Networks Pillar — Primary CSW Domain

Pillar	Stage Transition	ZTMM Requirement	CSW Implementation	Coverage
Networks	Traditional → Initial	Traffic Visibility	Establish initial network traffic visibility	ADM deployed; all workload flows documented; scope hierarchy defined
Networks	Initial → Advanced	Micro-Segmentation	Enforce workload-level network segmentation	Allowlist policies enforced; default-deny inbound; encryption compliance
Networks	Advanced → Optimal	Dynamic Policy	Continuously validate and dynamically adjust network policies	Continuous ADM re-runs; API-driven policy updates; ML anomaly detection
Networks	Advanced → Optimal	Encryption Enforcement	Encrypt all network traffic between workloads	Detect/block non-encrypted flows; enforce TLS-only paths across all scopes
Apps & Workloads	Traditional → Initial	Workload Inventory	Establish inventory of all application workloads	CSW inventory provides real-time workload asset catalog across hybrid environments
Apps & Workloads	Initial → Advanced	Workload Identity	Enforce workload identity-based access controls	Process hash + OS fingerprint + network identity — not just IP address
Apps & Workloads	Advanced → Optimal	Continuous Validation	Continuously validate workload access footprint	90-day ADM re-runs; vulnerability-triggered policy review; drift detection
Data	Traditional →	Data Flow	Establish visibility into data	ADM reveals

	Initial	Visibility	movement	all data flow paths across on-prem and cloud environments
Data	Initial → Advanced	Data Access Logging	Log all access to sensitive data	Full flow + process telemetry on sensitive data scopes; forensic export
Data	Advanced → Optimal	Encryption Compliance	Enforce encryption for sensitive data in transit	Protocol-level flow analysis; block plaintext; enforce HTTPS/SFTP/LDAPS
Identity	Supporting	Identity-Verified Paths	Enforce access only through identity-verified paths	CSW enforces that only identity-verified systems (LDAPS/Kerberos) can reach sensitive workloads
Devices	Supporting	Device Fingerprinting	Identify and authenticate devices accessing resources	OS fingerprint + process hash provides workload-level device identity in CSW

ZTMM Deployment Roadmap

Phase 1 — Traditional to Initial (Weeks 1–4)

- Deploy CSW sensors on all on-prem workloads; connect cloud accounts via CSW cloud connectors
- Run Application Dependency Mapping (ADM) to establish complete traffic baseline
- Define scope hierarchy aligned to ZTMM pillar boundaries
- Enable flow telemetry retention (minimum 12 months)

Phase 2 — Initial to Advanced (Weeks 5–12)

- Build allowlist policies from ADM baseline — enforce least-privilege network access
- Run simulation mode for 2 weeks; validate no false positives before enforcement
- Enable enforcement mode on sensitive scopes first; expand progressively
- Detect and block non-encrypted flows; enforce HTTPS/SFTP/LDAPS across all scopes
- Configure anomaly detection and SIEM integration for continuous monitoring

Phase 3 — Advanced to Optimal (Month 4+)

- Implement 90-day ADM re-run cycle for continuous policy validation
- Integrate CSW API with SOAR for automated policy response to incidents
- Enable vulnerability-triggered policy review — critical CVEs tighten access automatically
- Conduct quarterly ZTMM maturity assessment against scorecard

ZTMM Assessment Evidence Package

Evidence Item	CSW Source	ZTMM Pillar	Maturity Stage
ADM traffic baseline	Investigate → ADM	Networks, Apps	Initial
Enforcement policy export	Defend → Policy Workspaces	Networks, Apps	Advanced
Encryption compliance report	Flow Search (protocol filter)	Networks, Data	Advanced
Anomaly detection log	Alerts → Dashboard	Networks	Advanced
Vulnerability report	Investigate → Vulnerability	Apps & Workloads	Advanced
Continuous ADM comparison	ADM re-run vs baseline	Networks, Apps	Optimal
API policy automation log	CSW API audit log	Networks	Optimal

Workload inventory export	Manage → Inventory	Devices, Apps	Initial
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